

LAB04VitorAugustoCosta267117

1)

```
library(readxl)
library(tidyverse)
```

2)

```
fname <- "./WDIEXCEL.xlsx"
file.exists(fname)
```

[1] TRUE

3)

```
countries <- read_excel(fname, sheet = "Country")
```

4)

```
países_escolhidos <- c("BRA", "ARG", "CHL", "COL", "MEX")

countries <- countries %>%
  filter(`Country Code` %in% países_escolhidos)
```

5)

```
WDI <- read_csv("WDIEXCEL-Data.csv")
## pra nao estourar a memoria do positcloud, no chunk5 eu transformei os dados
## excel em csv e peguei apenas a DATA pra conseguir fazer o exercicio, se nao
## ele nao rodava nem com reza.
WDI <- WDI %>%
  filter(`Country Code` %in% paises_escolhidos) %>%
  filter(`Indicator Code` == "SP.DYN.CBRT.IN")
```

6)

```
WDI <- WDI %>%
  select(`Country Code`, `Indicator Code`, as.character(1960:2018))

### tem que usar crase quando tem espaco no nome das variaveis
```

7)

```
WDI <- WDI %>%
  pivot_longer(
    cols = as.character(1960:2018),
    names_to = "YEAR",
    values_to = "SP.DYN.CBRT.IN"
  ) %>%
  mutate(YEAR = as.integer(YEAR))
```

8)

```
WDI <- WDI %>%
  mutate(YEAR = as.integer(YEAR))
```

9)

```
WDI <- WDI %>%  
  left_join(countries, by = "Country Code") %>%  
  select(`Country Code`, `Short Name`, YEAR, SP.DYN.CBRT.IN)
```

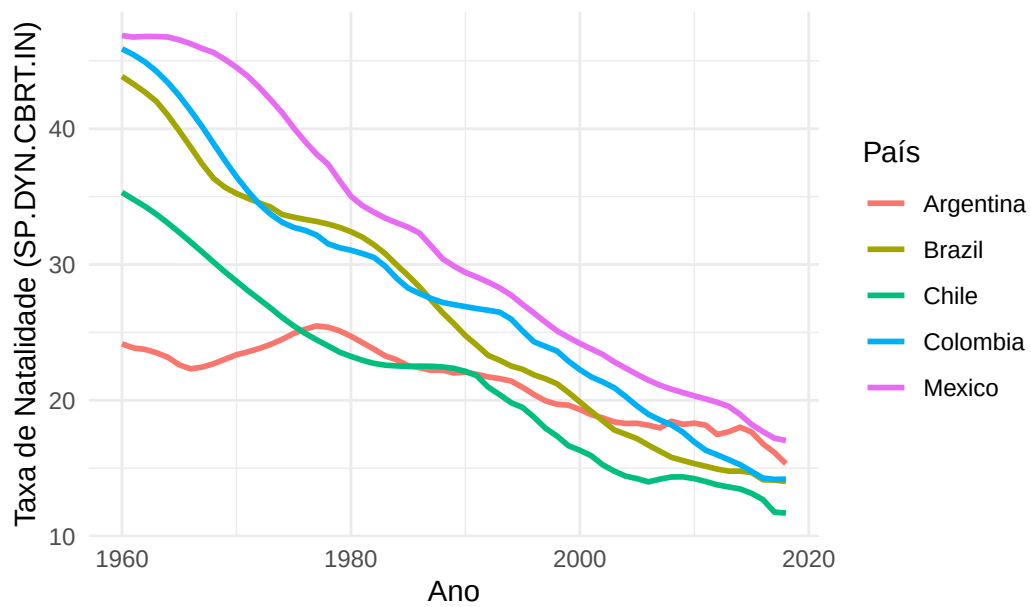
10)

```
WDI <- WDI %>%  
  arrange(`Short Name`, YEAR)
```

11)

```
ggplot(WDI, aes(x = YEAR, y = SP.DYN.CBRT.IN, color = `Short Name`)) +  
  geom_line(linewidth = 1) +  
  labs(  
    title = "Evolução da Taxa de Natalidade (1960-2018)",  
    x = "Ano",  
    y = "Taxa de Natalidade (SP.DYN.CBRT.IN)",  
    color = "País"  
  ) +  
  theme_minimal()
```

Evolução da Taxa de Natalidade (1960–2018)



horario

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-10 01:17:55 -03"
```